Exercise 3 : SQL Foundations - SQL Case Statements

1.
```
SELECT product_name, price,
CASE
WHEN price > 1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
WHEN price < 100 THEN 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2.
```
SELECT customer_name, amount,
CASE
WHEN amount >= 1000 THEN 'High value'
WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium-Value'
WHEN amount < 500 THEN 'Low Value'
END AS order_value_category
FROM orders;
```

Customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium ~~Low~~ Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

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3. SELECT emp_name, department, salary,
CASE
WHEN department = 'IT' AND Salary > 8000
~~WHEN~~ THEN 'Senior IT'
WHEN department = 'HR' AND Salary > 55000
THEN 'Experienced HR'
ELSE 'Staff'
END AS position-level
FROM employees;

emp-name	department	salary	position-level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	finance	95 000	Staff
Tom	HR	55 000	Staff

4. SELECT Student-name, score,
CASE
WHEN score ≥ 90 THEN 'A'
WHEN score ≥ 80 THEN 'B'
WHEN score ≥ 70 THEN 'C'
WHEN score ≥ 60 THEN 'D'
END AS grade
FROM students;

Student-name	score	grade
Anna	92	A
Ben	76	C
Cara	59	f
David	83	B
Ella	68	D

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5. SELECT delivery_id, delivery_time_minutes,
CASE
WHEN delivery_time_minutes ≤ 30 THEN 'Fast'
WHEN delivery_time_minutes ≤ 60 THEN 'On time'
ELSE 'Late'
END AS performance
FROM deliveries;

Delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	fast
4	65	Late
5	100	Late

6. SELECT issue_type, priority
CASE
~~WHERE~~ WHEN priority = 3 THEN 'High'
WHEN priority = 2 THEN 'Medium'
WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

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7. SELECT Student_id, ROUND((days_present x100/total_day
,2) AS attendence_percentage;
CASE
WHEN (days_present x100/total_days) ≥ 90
THEN 'Excellent'
WHEN (days_present x 100/total_days) ≥75
THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;

Student_id	attendance_percatge	atterdance_status
1	90.00	Excellent
2	60.00	Needs Improvement
3	96.00	Excellent
4	50.00	Needs Improvement
5	100,00	Excellent

8. SELECT product_id, stock_qty,
CASE
WHEN stock_qty =0 THEN
WHEN Stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
WHEN Stock_qty > 5 THEN 'In stock'
END AS Stock_status
FROM product_inventory;

product_id	Stock_qty	Stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

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9. SELECT subject, enrolled_students,
CASE
WHEN enrolled_students >= 25 THEN 'Large'
WHEN enrolled_students >= 10 THEN 'Medium'
ELSE 'Small'
END AS class_size_category
FROM classes;

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. SELECT payment_id, payment_method, amount
CASE
WHEN payment_method = 'Cash' AND amount >= 200
THEN 'Eligible for Discount'
ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not eligible
2	Cash	200.00	Eligible for discount
3	Card	150.00	Not eligible
4	Paypal	75.00	Not eligible
5	Cash	300.00	Eligible for discount

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